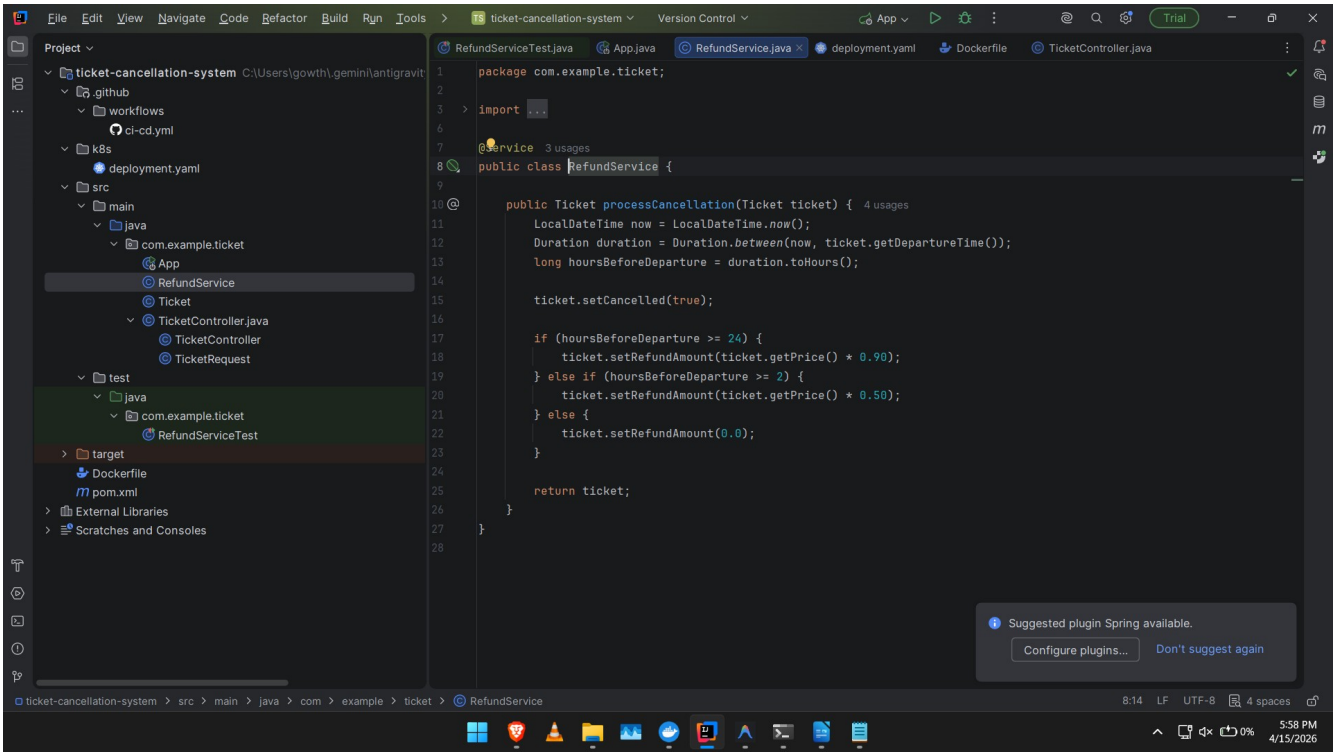
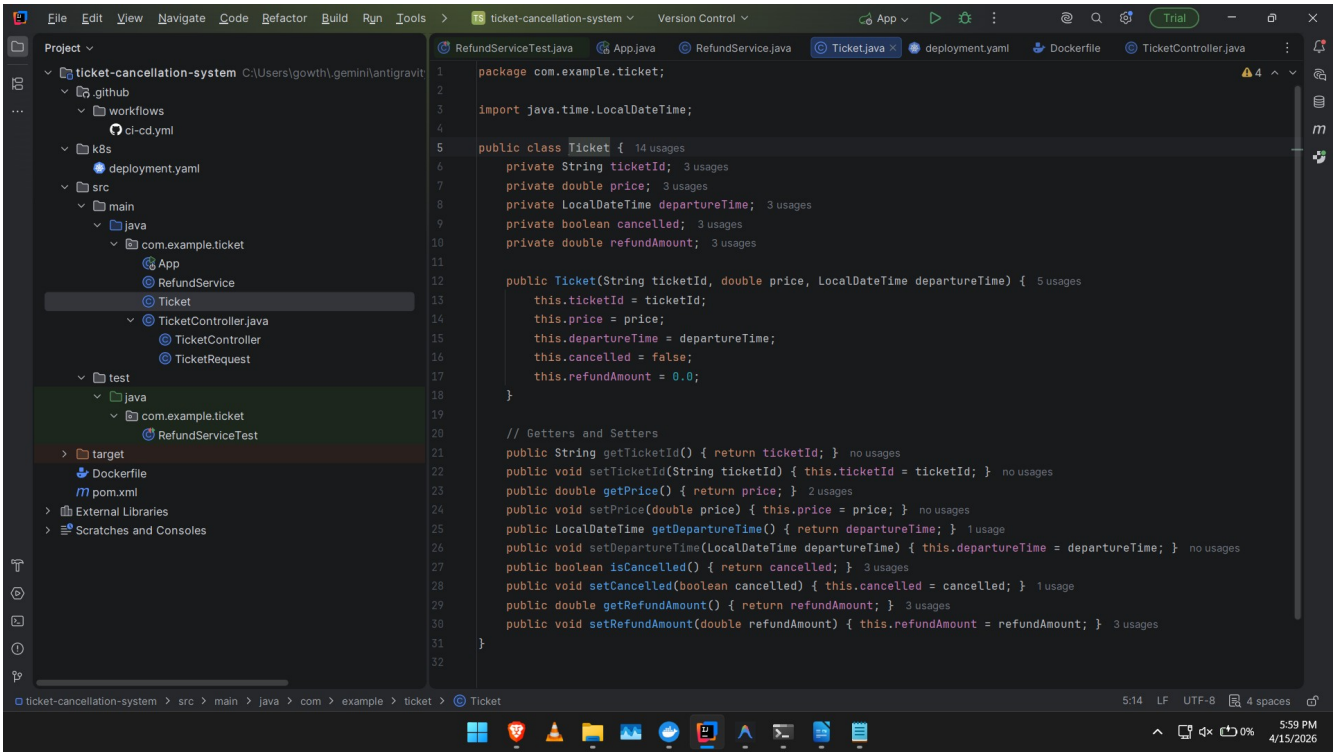
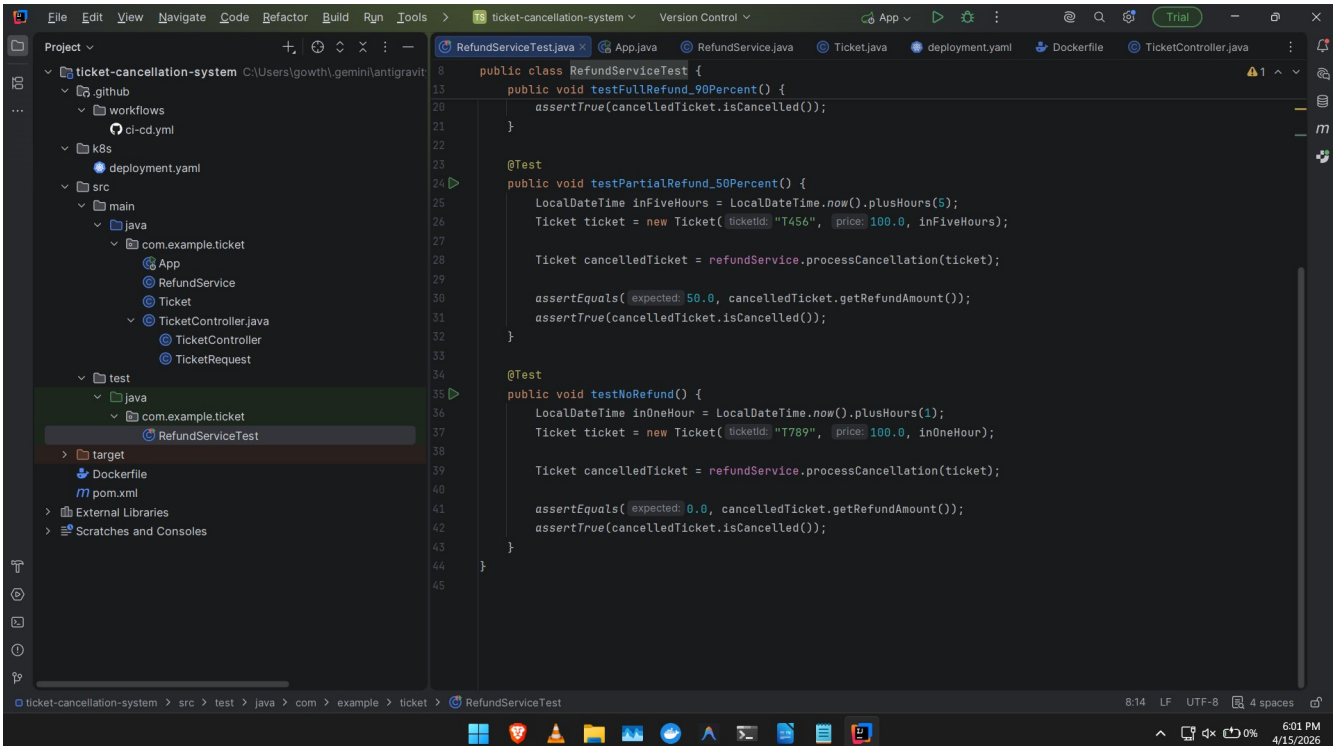
23MIS0336
GOWTHAM S
AGILE DEVELOPMENT PROCESS AND DEVOPS LAB
FACULTY:Dr.N.BALA GANESH
SET 3

Online Ticket Cancellation System
Problem Statement:
Ensure tickets are cancelled and refunded correctly.
Tasks:
1. Develop cancellation and refund logic

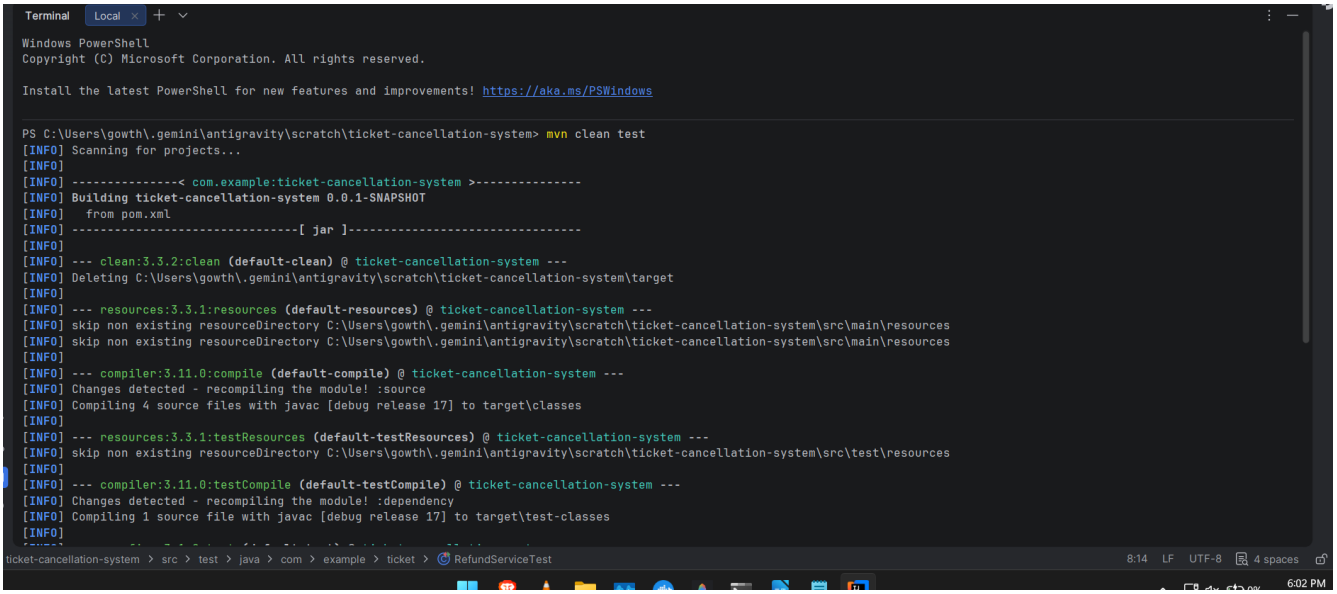


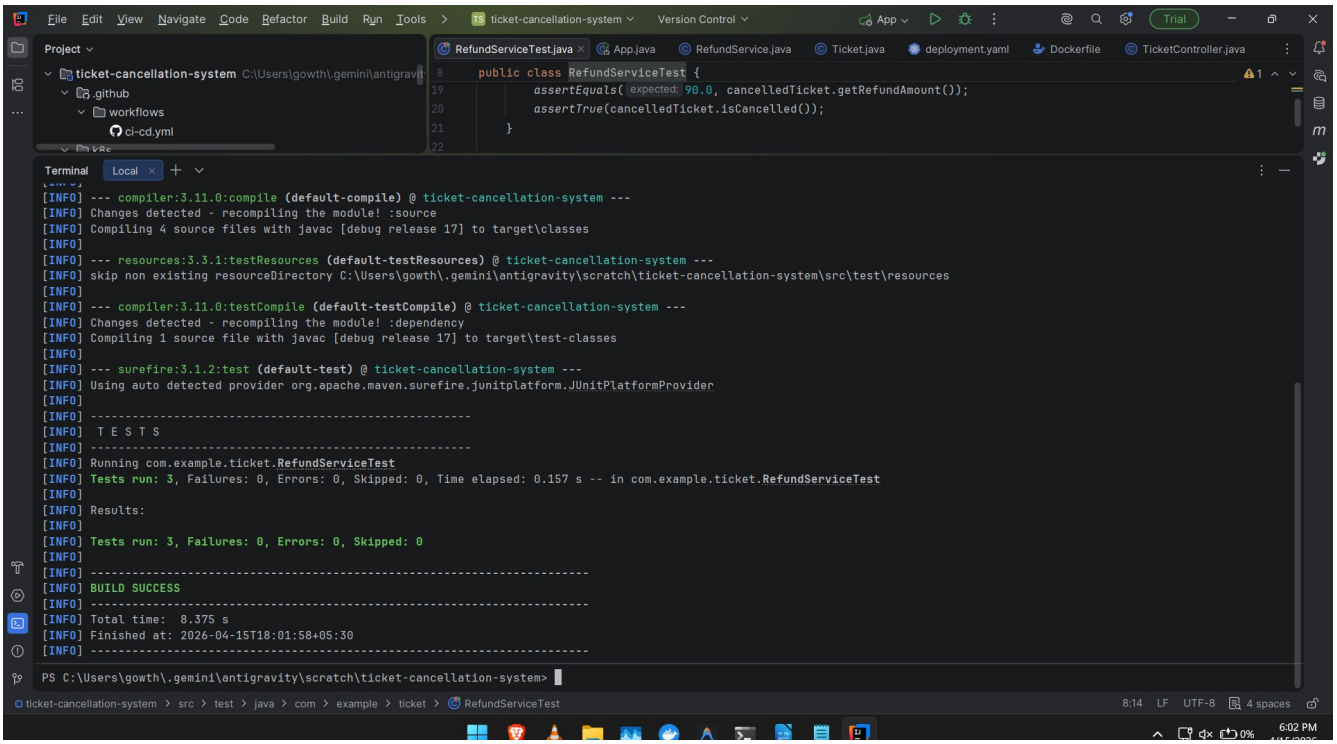


2. Write JUnit tests for refund scenarios



output

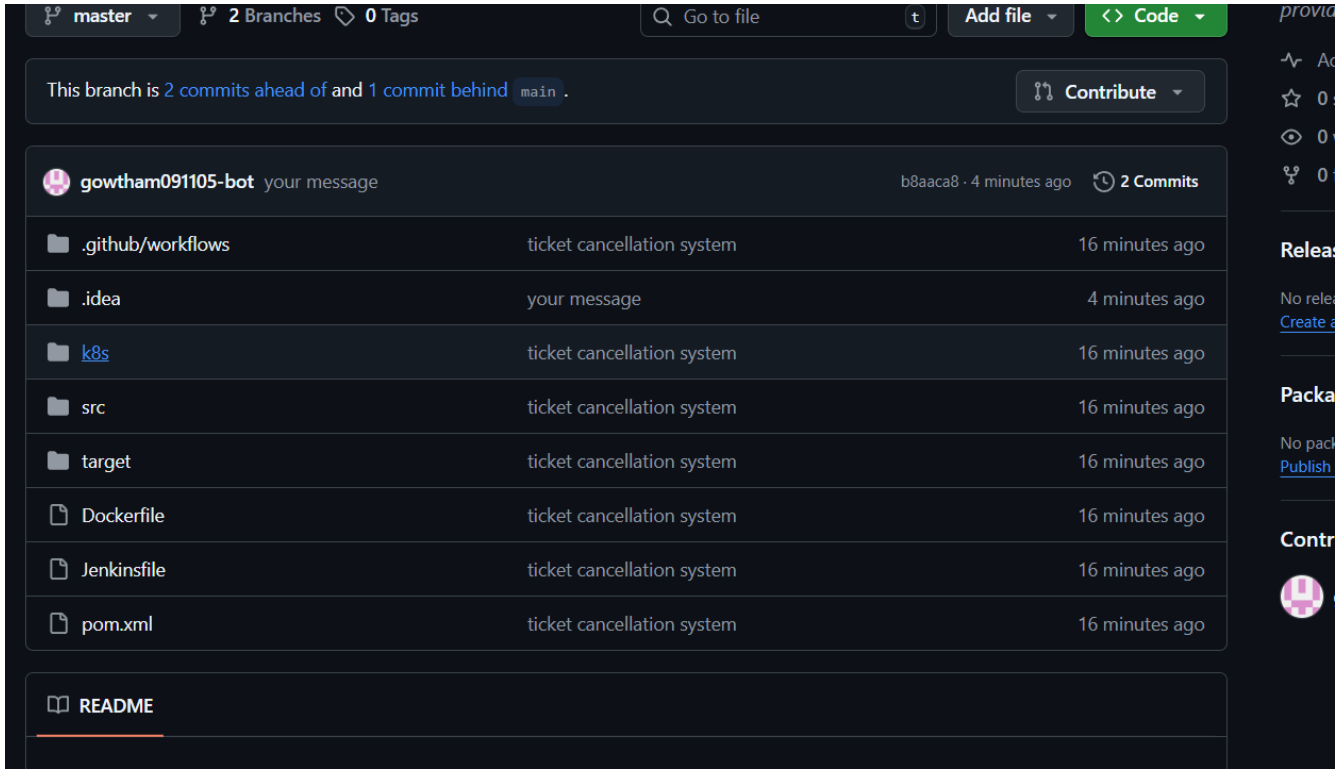




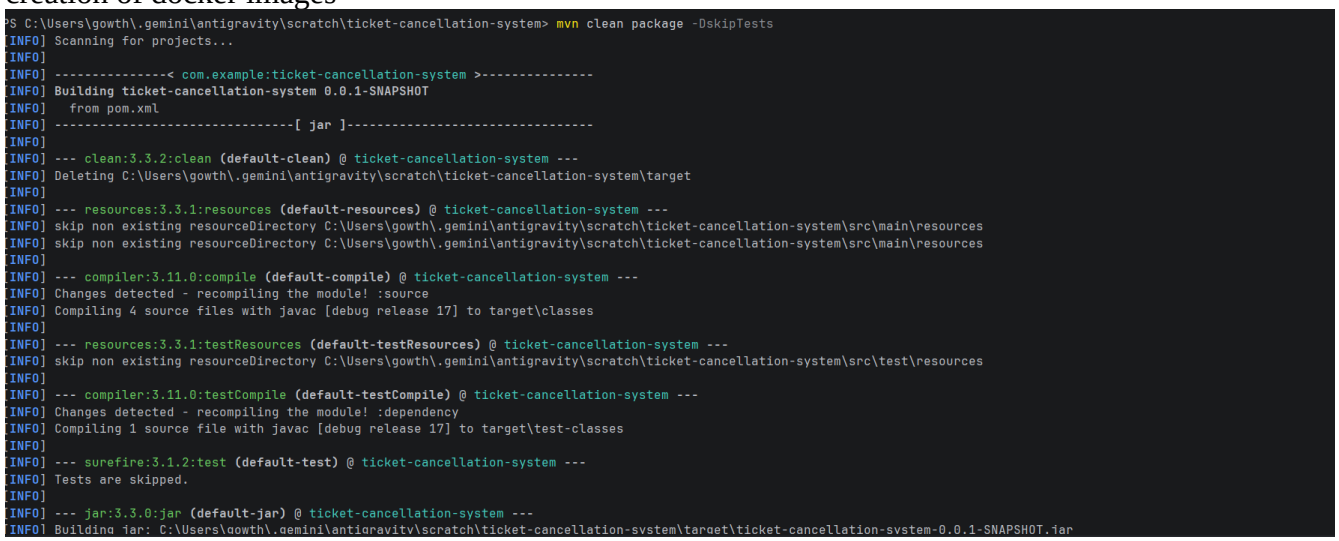
pushing code to github

```
Terminal   Local

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system> git init
Initialized empty Git repository in C:/Users/gowth/.gemini/antigravity/scratch/ticket-cancellation-system/.git/

PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system> git add .
warning: in the working copy of '.github/workflows/ci-cd.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Jenkinsfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'pom.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/ticket/App.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/ticket/RefundService.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/ticket/Ticket.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/ticket/TicketController.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/test/java/com/example/ticket/RefundServiceTest.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches i
t
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/surefire-reports/TEST-com.example.ticket.RefundServiceTest.xml', LF will be replaced by CRLF the next time Git touches it

PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system> git commit -m "ticket cancellation system"
[master (root-commit) 94c7c4b] ticket cancellation system
 27 files changed, 486 insertions(+)
 create mode 100644 .github/workflows/ci-cd.yml
 create mode 100644 .idea/.gitignore
 create mode 100644 .idea/compiler.xml
 create mode 100644 .idea/encodings.xml
 create mode 100644 .idea/jarRepositories.xml
 create mode 100644 .idea/misc.xml
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile

cket-cancellation-system > Jenkinsfile                    16:32  LF  UTF-8  4 spaces
                                                                                  6:17 PM
```

```
PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system> git push -u origin master
Enumerating objects: 60, done.
Counting objects: 100% (60/60), done.
Delta compression using up to 4 threads
Compressing objects: 100% (39/39), done.
Writing objects: 100% (60/60), 14.22 KiB | 297.00 KiB/s, done.
Total 60 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/gowtham091105-bot/ticket-cancellation-system/pull/new/master
remote:
To https://github.com/gowtham091105-bot/ticket-cancellation-system.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system>

ket-cancellation-system > Jenkinsfile                     16:32  LF  UTF-8  4 spaces
                                                                                  6:18 PM
```



creation of docker images



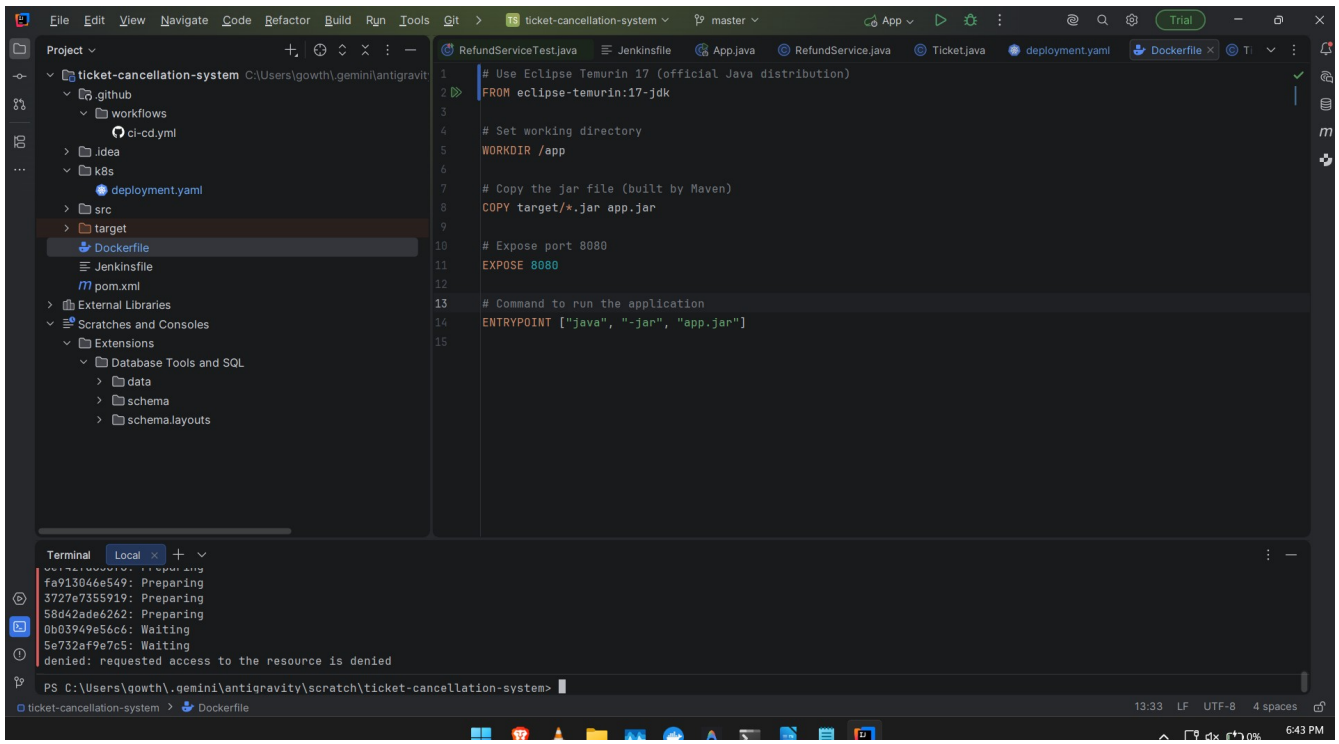
```
PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system> mvn clean package -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------< com.example:ticket-cancellation-system >---------------
[INFO] Building ticket-cancellation-system 0.0.1-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.3.2:clean (default-clean) @ ticket-cancellation-system ---
[INFO] Deleting C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ ticket-cancellation-system ---
[INFO] skip non existing resourceDirectory C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\src\main\resources
[INFO] skip non existing resourceDirectory C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ ticket-cancellation-system ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 4 source files with javac [debug release 17] to target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ ticket-cancellation-system ---
[INFO] skip non existing resourceDirectory C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ ticket-cancellation-system ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug release 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ ticket-cancellation-system ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ ticket-cancellation-system ---
[INFO] Building jar: C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\target\ticket-cancellation-system-0.0.1-SNAPSHOT.jar
```

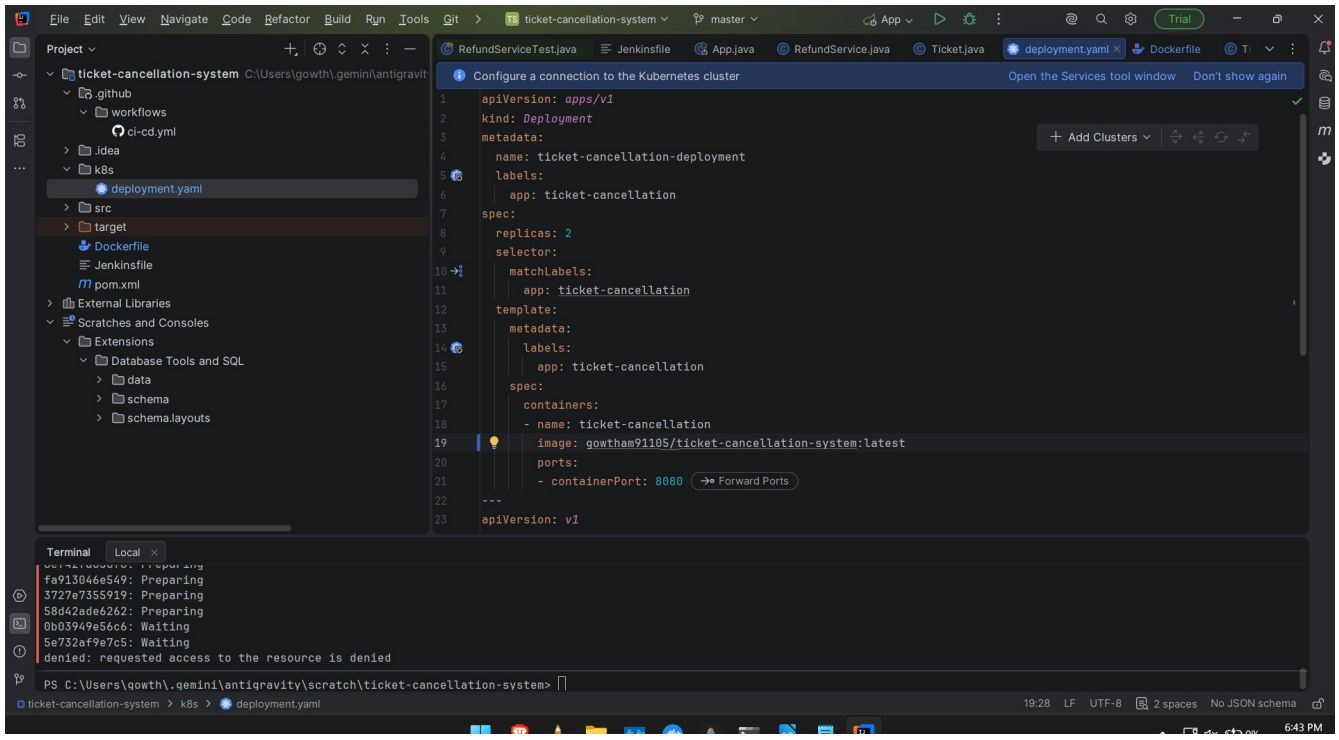
```
[INFO] --- compiler:3.11.0:compile (default-compile) @ ticket-cancellation-system ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 4 source files with javac [debug release 17] to target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ ticket-cancellation-system ---
[INFO] skip non existing resourceDirectory C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ ticket-cancellation-system ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug release 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ ticket-cancellation-system ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ ticket-cancellation-system ---
[INFO] Building jar: C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\target\ticket-cancellation-system-0.0.1-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:3.2.4:repackage (repackage) @ ticket-cancellation-system ---
[INFO] Replacing main artifact C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\target\ticket-cancellation-system-0.0.1-SNAPSHOT.jar with repackaged archive
ding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system\target\ticket-cancellation-system-0.0.1-SNAPSHOT.jar.origin
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.995 s
[INFO] Finished at: 2026-04-15T18:35:12+05:30
[INFO] ------------------------------------------------------------------------
```

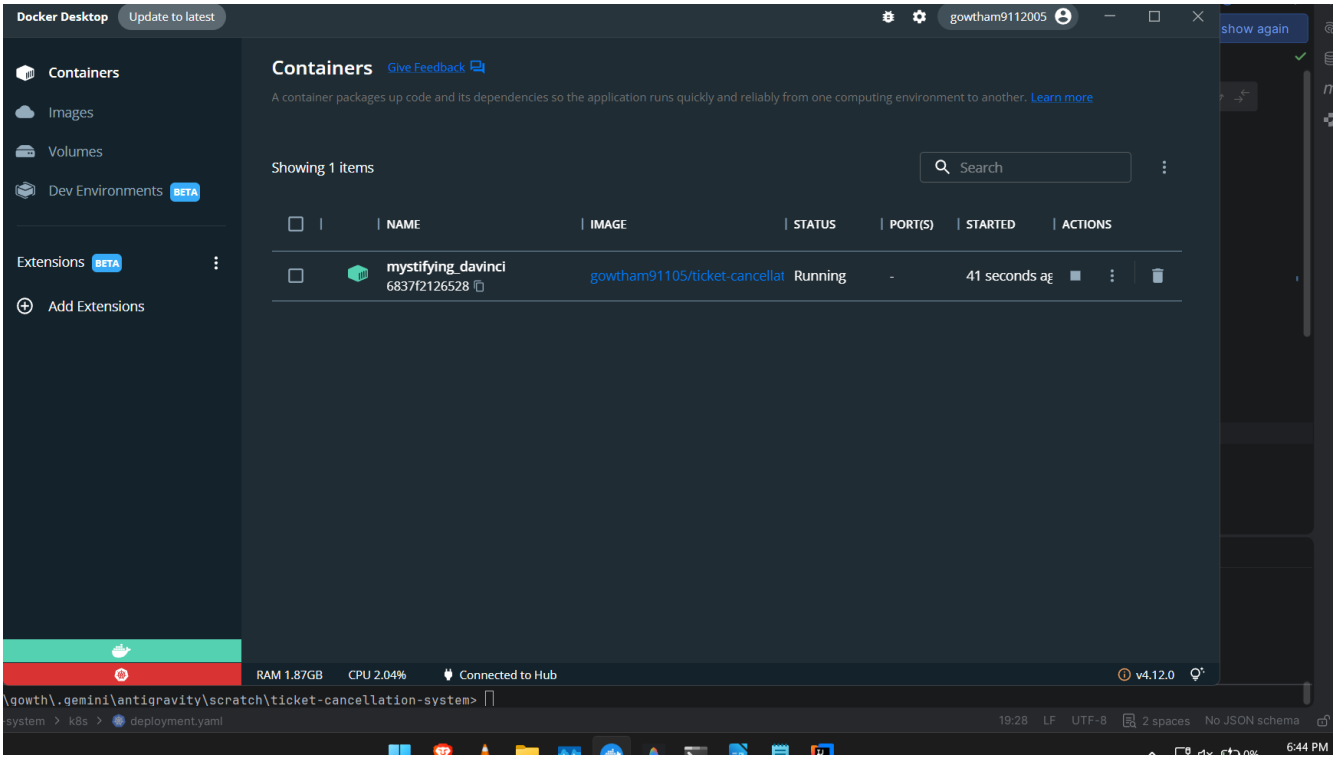
building process

```
PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system> docker build -t gowtham91105/ticket-cancellation-system:latest .
[+] Building 33.4s (5/8)
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 328B
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io
 => [internal] load build context
 => => transferring context: 19.76MB
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430
 => => sha256:a7428f4f5c1dad835231adfcff057cad6efc5b2dc256a36871e2e72cb75561e4 114.29MB / 145.63MB
 => => sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c99bcf1430 8.48kB / 8.48kB
 => => sha256:884c0b17d32c2889ce4cf820a1c0238ef97ea47bf8a53785959ce3617f7d63df 1.94kB / 1.94kB
 => => sha256:ddb0d3a563764715968e4428766e2e66a9b6b6550829c0f3ef3d31c76f24eb77 6.28kB / 6.28kB
 => => sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6 29.73MB / 29.73MB
 => => sha256:a1e6e1a167f1d8e27a6ca55c32fa609912f9ff5b0fc88c77413259b59e72ac45 22.97MB / 22.97MB
 => => sha256:a149c623348dfa371df99c86694ffdf860874791706585045a923ea4e6224b0d 158B / 158B
 => => sha256:f00bd8c488bba663bd5f60752b0a1ac92bc39a10729a5a8bf7227b10f6096904 2.28kB / 2.28kB
 => => extracting sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6
```

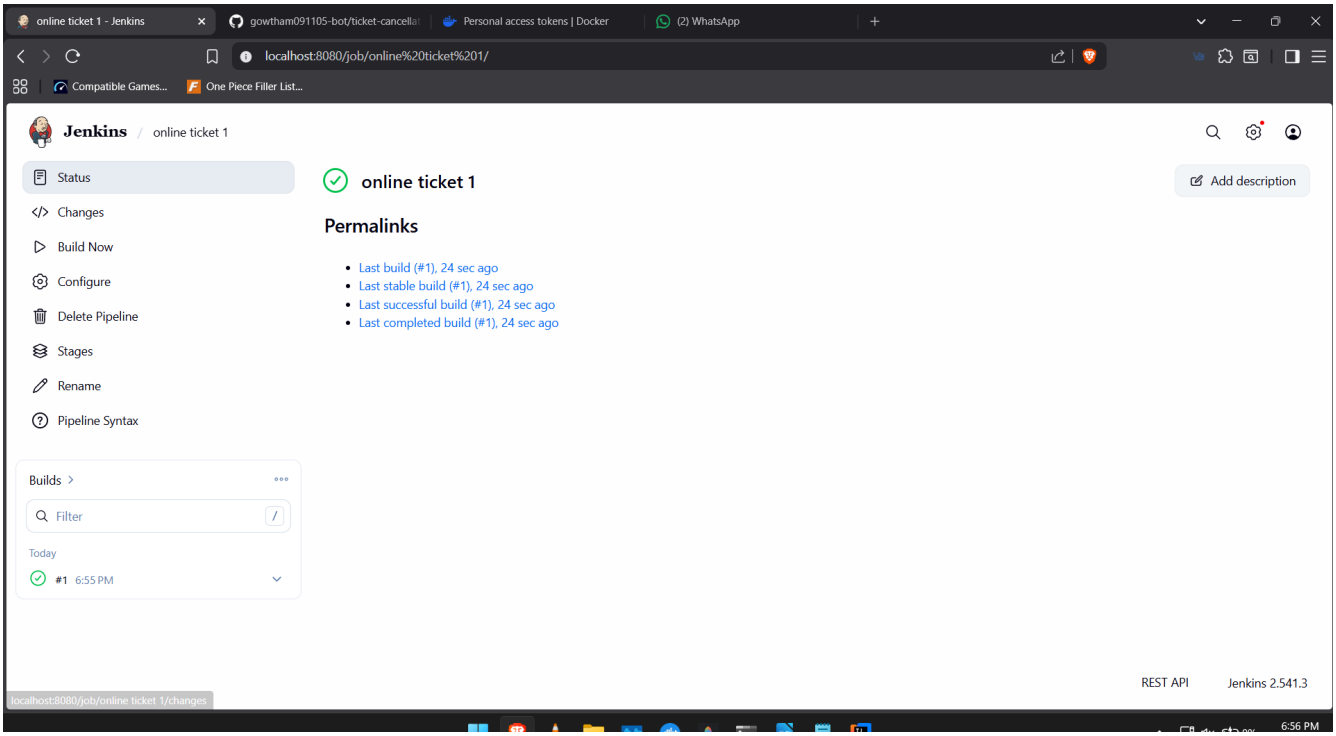
docker and kubernetes images and yaml codes


File Edit View Navigate Code Refactor Build Run Tools Git
ticket-cancellation-system
master
App
Trial
Project
ticket-cancellation-system C:\Users\gowth\.gemini\antigravit
.github
workflows
ci-cd.yml
.idea
k8s
deployment.yaml
src
target
Dockerfile
Jenkinsfile
pom.xml
External Libraries
Scratches and Consoles
Extensions
Database Tools and SQL
data
schema
schema.layouts
RefundServiceTest.java
Jenkinsfile
App.java
RefundService.java
Ticket.java
deployment.yaml
Dockerfile
Use Eclipse Temurin 17 (official Java distribution)
FROM eclipse-temurin:17-jdk
Set working directory
WORKDIR /app
Copy the jar file (built by Maven)
COPY target/*.jar app.jar
Expose port 8080
EXPOSE 8080
Command to run the application
ENTRYPOINT ["java", "-jar", "app.jar"]
Terminal
Local
fa913046e549: Preparing
3727e7355919: Preparing
58d42ade6262: Preparing
0b03949e56c6: Waiting
5e732af9e7c5: Waiting
denied: requested access to the resource is denied
PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system>
ticket-cancellation-system > Dockerfile
13:33 LF UTF-8 4 spaces
6:43 PM


File Edit View Navigate Code Refactor Build Run Tools Git
ticket-cancellation-system
master
App
Trial
Project
ticket-cancellation-system C:\Users\gowth\.gemini\antigravit
.github
workflows
ci-cd.yml
.idea
k8s
deployment.yaml
src
target
Dockerfile
Jenkinsfile
pom.xml
External Libraries
Scratches and Consoles
Extensions
Database Tools and SQL
data
schema
schema.layouts
RefundServiceTest.java
Jenkinsfile
App.java
RefundService.java
Ticket.java
deployment.yaml
Dockerfile
Configure a connection to the Kubernetes cluster
Open the Services tool window
Don't show again
Add Clusters
apiVersion: apps/v1
kind: Deployment
metadata:
name: ticket-cancellation-deployment
labels:
app: ticket-cancellation
spec:
replicas: 2
selector:
matchLabels:
app: ticket-cancellation
template:
metadata:
labels:
app: ticket-cancellation
spec:
containers:
- name: ticket-cancellation
image: gowtham91105/ticket-cancellation-system:latest
ports:
- containerPort: 8080
Forward Ports
---
apiVersion: v1
Terminal
Local
fa913046e549: Preparing
3727e7355919: Preparing
58d42ade6262: Preparing
0b03949e56c6: Waiting
5e732af9e7c5: Waiting
denied: requested access to the resource is denied
PS C:\Users\gowth\.gemini\antigravity\scratch\ticket-cancellation-system>
ticket-cancellation-system > k8s > deployment.yaml
19:28 LF UTF-8 2 spaces No JSON schema
6:43 PM



ci cd pipelines in git and in jenkins



output

```
[Pipeline] }
[Pipeline] // catchError
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Final Status)
[Pipeline] echo
Pipeline executed successfully
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.3

7:04 PM

S	W	Name ↓	Last Success	Last Failure	Last Duration	
✓	☼	online ticket 1	1 min 57 sec #1	N/A	16 sec	▷